

Class 10 Science – Chapter 1:

Chemical Reactions and Equations.

1. The process in which a substance reacts with oxygen to give heat and light is called:

- A) Decomposition reaction
- B) Oxidation reaction
- C) Combustion reaction
- D) Neutralisation reaction

Answer: C) Combustion reaction

2. Which of the following represents a balanced chemical equation?

- A) $\text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{O}$
- B) $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$
- C) $\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}_2$
- D) $2\text{H}_2\text{O} \rightarrow 2\text{H}_2 + \text{O}_2$

Answer: B) $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$

3. The chemical reaction between quicklime and water to form slaked lime is:

- A) Endothermic
- B) Exothermic
- C) Displacement
- D) Double displacement

Answer: B) Exothermic

4. Which of the following is a decomposition reaction?

- A) $\text{Zn} + \text{H}_2\text{SO}_4 \rightarrow \text{ZnSO}_4 + \text{H}_2$
- B) $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$
- C) $2\text{Pb}(\text{NO}_3)_2 \rightarrow 2\text{PbO} + 4\text{NO}_2 + \text{O}_2$
- D) $\text{NaOH} + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O}$

Answer: C) $2\text{Pb}(\text{NO}_3)_2 \rightarrow 2\text{PbO} + 4\text{NO}_2 + \text{O}_2$

5. The brown gas evolved on heating lead nitrate is:

- A) NO_2
- B) N_2O
- C) NO
- D) O_2

Answer: A) NO_2

6. The black coating on silver articles is due to:

- A) Carbon
- B) Sulphur compounds in air
- C) Hydrogen sulphide gas
- D) Oxidation

Answer: C) Hydrogen sulphide gas

7. $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$ is an example of:

- A) Combination reaction
- B) Double displacement reaction
- C) Displacement reaction
- D) Decomposition reaction

Answer: C) Displacement reaction

8. The product formed when calcium oxide reacts with water is:

- A) Ca(OH)_2
- B) CaCO_3
- C) CaCl_2
- D) CaSO_4

Answer: A) Ca(OH)_2

9. The equation $2\text{KClO}_3 \rightarrow 2\text{KCl} + 3\text{O}_2$ is an example of:

- A) Combination reaction
- B) Decomposition reaction
- C) Neutralisation
- D) Double displacement

Answer: B) Decomposition reaction

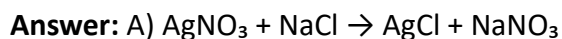
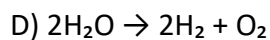
10. The color of copper sulphate solution changes when iron is dipped in it because:

- A) Cu^{2+} is displaced by Fe
- B) Fe^{2+} is displaced by Cu
- C) Both metals form complex
- D) None of these

Answer: A) Cu^{2+} is displaced by Fe

11. Which of the following is a double displacement reaction?

- A) $\text{AgNO}_3 + \text{NaCl} \rightarrow \text{AgCl} + \text{NaNO}_3$
- B) $2\text{KClO}_3 \rightarrow 2\text{KCl} + 3\text{O}_2$
- C) $\text{H}_2 + \text{Cl}_2 \rightarrow 2\text{HCl}$



12. The reaction $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$ is:

- A) Combination
- B) Decomposition
- C) Displacement
- D) Redox

Answer: A) Combination

13. The reaction in which heat is absorbed is called:

- A) Exothermic
- B) Endothermic
- C) Neutralisation
- D) Displacement

Answer: B) Endothermic

14. Corrosion of iron is commonly known as:

- A) Tarnishing
- B) Rusting
- C) Decomposition
- D) Redox

Answer: B) Rusting

15. Which gas is released when zinc reacts with dilute sulphuric acid?

- A) Oxygen
- B) Hydrogen
- C) Carbon dioxide
- D) Nitrogen

Answer: B) Hydrogen

16. The chemical formula of quicklime is:

- A) CaCO_3
- B) Ca(OH)_2
- C) CaO
- D) CaCl_2

Answer: C) CaO

17. In $\text{Zn} + \text{H}_2\text{SO}_4 \rightarrow \text{ZnSO}_4 + \text{H}_2$, zinc acts as:

- A) Oxidising agent
- B) Reducing agent
- C) Catalyst
- D) None

Answer: B) Reducing agent

18. When ferrous sulphate is heated, the residue left is:

- A) FeS
- B) Fe_2O_3
- C) FeO
- D) Fe_3O_4

Answer: B) Fe_2O_3

19. The process of depositing a thin layer of zinc on iron objects is called:

- A) Corrosion
- B) Rusting
- C) Galvanisation
- D) Electroplating

Answer: C) Galvanisation

20. Which one of the following is not a chemical change?

- A) Digestion of food
- B) Melting of ice
- C) Rusting of iron
- D) Burning of candle

Answer: B) Melting of ice

21. The reaction $2\text{HgO} \rightarrow 2\text{Hg} + \text{O}_2$ is an example of:

- A) Combination
- B) Decomposition
- C) Double displacement
- D) Displacement

Answer: B) Decomposition

22. In $\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$, copper is:

- A) Oxidised
- B) Reduced
- C) Neither

D) Both

Answer: B) Reduced

23. The reaction between acid and base to form salt and water is called:

A) Displacement reaction

B) Neutralisation reaction

C) Combination reaction

D) Redox reaction

Answer: B) Neutralisation reaction

24. What happens when magnesium ribbon is heated in air?

A) It burns with a brilliant white light

B) It turns brown

C) It forms MgCl_2

D) It emits no light

Answer: A) It burns with a brilliant white light

25. A reaction in which oxidation and reduction occur simultaneously is called:

A) Combination

B) Displacement

C) Redox

D) Double displacement

Answer: C) Redox

26. The green coating formed on copper over time is due to formation of:

A) Basic copper carbonate

B) CuO

C) CuSO_4

D) CuCl_2

Answer: A) Basic copper carbonate

27. The symbol (aq) in a chemical equation stands for:

A) Aqueous solution

B) Air

C) Solid

D) Gas

Answer: A) Aqueous solution

28. The chemical change in which two compounds exchange ions is called:

- A) Decomposition
- B) Double displacement
- C) Combination
- D) Neutralisation

Answer: B) Double displacement

29. Which reaction produces a precipitate?

- A) $\text{AgNO}_3 + \text{NaCl} \rightarrow \text{AgCl} + \text{NaNO}_3$
- B) $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$
- C) $\text{Zn} + \text{H}_2\text{SO}_4 \rightarrow \text{ZnSO}_4 + \text{H}_2$
- D) $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2$

Answer: A) $\text{AgNO}_3 + \text{NaCl} \rightarrow \text{AgCl} + \text{NaNO}_3$

30. The substance oxidised in the reaction $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$ is:

- A) Mg
- B) O_2
- C) MgO
- D) None

Answer: A) Mg
